



Place of articulation

In articulatory phonetics, the **place of articulation** (also **point of articulation**) of a consonant is an approximate location along the vocal tract where its production occurs.^{[1]:10} It is a point where a constriction is made between an active and a passive articulator. Active articulators are organs capable of voluntary movement which create the constriction, while passive articulators are so called because they are normally fixed and are the parts with which an active articulator makes contact.^{[2]:24} Along with the manner of articulation and phonation, the place of articulation gives the consonant its distinctive sound.

Since vowels are produced with an open vocal tract, the point where their production occurs cannot be easily determined. Therefore, they are not described in terms of a place of articulation but by the relative positions in vowel space. This is mostly dependent on their formant frequencies and less on the specific tongue position and lip rounding.^{[3]:34}

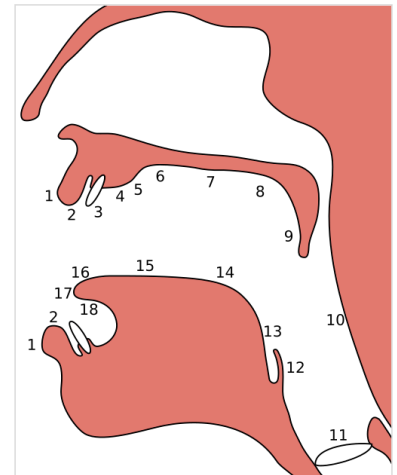
The terminology used in describing places of articulation has been developed to allow specifying of all theoretically possible contrasts. No known language distinguishes all of the places described in the literature so less precision is needed to distinguish the sounds of a particular language.^{[1]:39}

Overview

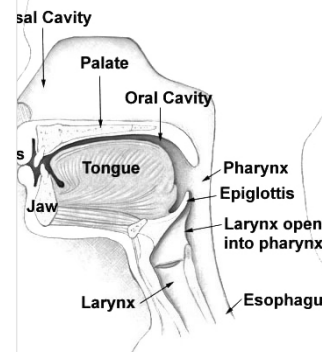
The human voice produces sounds in the following manner:^{[4][5]}

1. Air pressure from the lungs creates a steady flow of air through the trachea (voice box) and pharynx (back of the throat). Therefore, the air moves through a coordinated action of the diaphragm, abdominal muscles, chest cage.
2. The vocal folds in the larynx vibrate, creating fluctuations in air pressure.
3. Resonances in the vocal tract modify these waves according to the positions of the speech organs, creating formant regions and so different qualities of sound.
4. Mouth radiates the sound waves into the environment.
5. Nasal cavity adds resonance to some sounds such as [m] and [n] to give them a nasal quality.

In articulatory phonetics, the **manner of articulation** is the configuration and interaction of the articulators when making a speech sound. One parameter of manner is *stricture*, that is, how closely the speech organs approach each other.



Common approximate places of articulation (passive & active):
1. Exo-labial, 2. Endo-labial, 3. Dental, 4. Alveolar, 5. Post-alveolar, 6. Pre-palatal, 7. Palatal, 8. Velar, 9. Uvular, 10. Pharyngeal, 11. Glottal, 12. Epiglottal, 13. Radical, 14. Postero-dorsal, 15. Antero-dorsal, 16. Laminal, 17. Apical, 18. Sub-



The larynx

The *larynx* or *voice box* is a cylindrical framework of cartilage that serves to anchor the vocal folds. When the muscles of the vocal folds contract, the airflow from the lungs is impeded until the vocal folds are forced apart again by the increasing air pressure from the lungs. The process continues in a periodic cycle that is felt as a vibration (buzzing). In singing, the vibration frequency of the vocal folds determines the pitch of the sound produced. Voiced phonemes such as the pure vowels are, by definition, distinguished by the buzzing sound of this periodic oscillation of the vocal cords.

The lips of the mouth can be used in a similar way to create a similar sound, as any toddler or trumpeter can demonstrate. A rubber balloon, inflated but not tied off and stretched tightly across the neck produces a squeak or buzz, depending on the tension across the neck and the level of pressure inside the balloon. Similar actions with similar results occur when the vocal cords are contracted or relaxed across the larynx.

Active articulators

The active articulators are movable parts of the vocal apparatus that impede or direct the airstream, typically some part of the tongue or lips.^{[3]:4} There are five major parts of the vocal tract that move: the lips, the flexible front of the tongue, the body of the tongue, the root of the tongue together with the epiglottis, and the glottis. They are discrete in that they can act independently of each other, and two or more may work together in what is called *coarticulation*.^{[1]:10-11}

The five main active parts can be further divided, as many languages contrast sounds produced within the same major part of the vocal apparatus. The following 9 degrees of active articulatory areas are known to be contrastive (sorted such that the top-most is in the front-most area of the mouth and the bottom-most is in the rear-most area of the mouth):^{[1]:10-15}

- The lower lip (*labial*)
- Various parts of the front of the tongue (*coronal*):
 - The tip of the tongue (*apical*)
 - The upper front surface of the tongue just behind the tip, called the *blade* of the tongue (*laminal*)
 - The surface of the tongue *under* the tip (*subapical*)
- The body of the tongue (*dorsal*) which is sometimes further divided into front and back
- The base a.k.a. root of the tongue and the throat (*pharyngeal*)
- The aryepiglottic fold inside the throat (*aryepiglottal*)
- The glottis at the very back of the windpipe (*glottal*)

In bilabial consonants, both lips move so the articulatory gesture brings the lips together, but by convention, the lower lip is said to be active and the upper lip passive. Similarly, in linguolabial consonants the tongue contacts the upper lip with the upper lip actively moving down to meet the tongue; nonetheless, the tongue is conventionally said to be active and the lip passive if for no other reason than that the parts of the mouth below the vocal tract are typically active, and those above the vocal tract are typically passive.

In dorsal gestures, different parts of the body of the tongue contact different parts of the roof of the mouth, but it cannot be independently controlled so they are all subsumed under the term *dorsal*. That is unlike coronal gestures involving the front of the tongue, which is more flexible.

The epiglottis may be active, contacting the pharynx, or passive, being contacted by the aryepiglottal folds. Distinctions made in these laryngeal areas are very difficult to observe and are the subject of ongoing investigation, and several still-unidentified combinations are thought possible.

The glottis acts upon itself. There is a sometimes fuzzy line between glottal, aryepiglottal, and epiglottal consonants and phonation, which uses these same areas.

Passive articulators

The passive are the more stationary parts of the vocal tract that the active articulator touches or gets close to; they can be anywhere from the lips, upper teeth, gums, or roof of the mouth to the back of the throat.^{[3]:4} Although it is a continuum, there are several contrastive areas so languages may distinguish consonants by articulating them in different areas, but few languages contrast two sounds within the same area unless there is some other feature which contrasts as well. The following 9 degrees of passive articulatory areas are known to be contrastive (sorted such that the top-most is in the front-most area of the mouth and the bottom-most is in the rear-most area of the mouth):

- The upper lip (*labial*)
- The upper teeth, either on the edge of the teeth or inner surface (*dental*)
- The alveolar ridge, the gum line just behind the teeth (*alveolar*)
- The back of the alveolar ridge (*post-alveolar*)
- The hard palate on the roof of the mouth (*palatal*)
- The soft palate further back on the roof of the mouth (*velar*)
- The uvula hanging down at the entrance to the throat (*uvular*)
- The throat itself, a.k.a. the pharynx (*pharyngeal*)
- The epiglottis at the entrance to the windpipe, above the voice box (*epiglottal*)

The regions are not strictly separated. For instance, in some sounds in many languages, the surface of the tongue contacts a relatively large area from the back of the upper teeth to the alveolar ridge, which is common enough to have received its own name, *denti-alveolar*. Likewise, the alveolar and post-alveolar regions merge into each other, as do the hard and soft palate, the soft palate and the uvula, and all adjacent regions. Terms like *pre-velar* (intermediate between palatal and velar), *post-velar* (between velar and uvular), and *upper* vs. *lower* pharyngeal may be used to specify more precisely where an articulation takes place. However, although a language may contrast pre-velar and post-velar sounds, it does not also contrast them with palatal and uvular sounds (of the same type of consonant) so contrasts are limited to the number above, if not always their exact location.

Table of gestures and passive articulators and resulting places of articulation

The following table shows the possible combinations of active and passive articulators.

The possible locations for sibilants as well as non-sibilants to occur are indicated in dashed red. For sibilants, there are additional complications involving tongue shape; see the article on sibilants for a chart of possible articulations.

	Front/back →	Front				Back		
	Major class →	Labial	Coronal			"Guttural"		
Acute/grave ↓	Active articulator →	Lower lip (Labial)	Tongue blade (Laminal)	Tongue tip (Apical)	Underside of tongue (Subapical)	Tongue body (Dorsal)	Tongue root (Radical)	Larynx (Laryngeal)
	Passive articulator ↓							
Grave	Upper lip	bilabial	linguolabial					
	Upper teeth	labiodental						
Acute	Upper teeth		interdental	dental				
	Upper teeth/alveolar ridge		denti-alveolar					
	Alveolar ridge		laminal alveolar	apico-alveolar				
	Back of alveolar ridge (postalveolar)		palato-alveolar	apical retroflex		alveolo-palatal		
	Hard palate (front)				retroflex	palatal		
Grave	Soft palate				subapical velar	velar		
	Uvula					uvular		
	Pharynx						pharyngeal	epiglottopharyngeal
	Epiglottis							(ary-)epiglottal
	Glottis							glottal

A precise vocabulary of compounding the two places of articulation is sometimes seen. However, it is usually reduced to the passive articulation, which is generally sufficient. Thus *dorsal–palatal*, *dorsal–velar*, and *dorsal–uvular* are usually just called "palatal", "velar", and "uvular". If there is ambiguity, additional terms have been invented, so *subapical–palatal* is more commonly called "retroflex".

Note: Additional shades of passive articulation are sometimes specified using *pre-* or *post-*, for example *prepalatal* (near the border between the postalveolar region and the hard palate; *prevelar* (at the back of the hard palate, also *post-palatal* or even *medio-palatal* for the middle of the hard palate); or *postvelar* (near the border of the soft palate and the uvula). They can be useful in the precise description of sounds that are articulated somewhat farther forward or back than a prototypical consonant; for this purpose, the "fronted" and "retracted" IPA diacritics can be used. However, no additional shade is needed to phonemically distinguish two consonants in a single language.^[a]

Homorganic consonants

Consonants that have the same place of articulation, such as the alveolar sounds /n, t, d, s, z, l/ in English, are said to be *homorganic*. Similarly, labial /p, b, m/ and velar /k, g, ŋ/ are homorganic. A homorganic nasal rule, an instance of assimilation, operates in many languages, where a nasal consonant must be homorganic with a following stop. We see this with English *intolerable* but *implausible*; another example is found in Yoruba, where the present tense of *ba* "hide" is *mba* "is hiding", while the present of *sun* "sleep" is *nsun* "is sleeping".

Central and lateral articulation

The tongue contacts the mouth with a surface that has two dimensions: length and width. So far, only points of articulation along its length have been considered. However, articulation varies along its width as well. When the airstream is directed down the center of the tongue, the consonant is said to be *central*. If, however, it is deflected off to one side, escaping between the side of the tongue and the side teeth, it is said to be *lateral*. Nonetheless, for simplicity's sake the place of articulation is assumed to be the point along the length of the tongue, and the consonant may in addition be said to be central or lateral. That is, a consonant may be lateral alveolar, like English /l/ (the tongue contacts the alveolar ridge, but allows air to flow off to the side), or lateral palatal, like Castilian Spanish *ll* /ʎ/. Some Indigenous Australian languages contrast dental, alveolar, retroflex, and palatal laterals, and many Native American languages have lateral fricatives and affricates as well.

Coarticulation

Some languages have consonants with two simultaneous places of articulation, which is called **coarticulation**. When these are doubly articulated, the articulators must be independently movable, and therefore there may be only one each from the major categories *labial*, *coronal*, *dorsal* and *pharyngeal*.

The only common doubly articulated consonants are **labial–velar** stops like [kp̚], [gb̚] and less commonly [ɲm̚], which are found throughout **Western Africa** and **Central Africa**. Other combinations are rare but include **labial–(post)alveolar stops** [tp̚ db̚ nm̚], found as distinct consonants only in a single language in New Guinea, and a **uvular–epiglottal stop**, [q̠̚], found in **Somali**.

More commonly, coarticulation involves **secondary articulation** of an **approximantic** nature. Then, both articulations can be similar such as labialized labial [mʷ] or palatalized velar [kʲ]. That is the case of English [w], which is a velar consonant with secondary labial articulation.

Common coarticulations include these:

- **Labialization**, rounding the lips while producing the obstruction, as in [kʷ] and English [w].
- **Palatalization**, raising the body of the tongue toward the hard palate while producing the obstruction, as in Russian [tʲ] and [ç].
- **Velarization**, raising the back of the tongue toward the soft palate (velum), as in the English dark el, [ɫ] (also transcribed [ɫ̠]).
- **Pharyngealization**, constriction of the throat (**pharynx**), such as **Arabic** "emphatic" [tˤ].

See also

- **Articulatory phonetics**
- **Manner of articulation**
- **Relative articulation**
- **Tongue shape**
- **Sibilant**
- **Index of phonetics articles**

Notes

- a. Occasionally claims to the contrary are met. For example, some dialects of Malayalam are said to distinguish palatal, prevelar and velar consonants. In reality, the dialects distinguish palato-alveolar (palatalized postalveolar), palatal and velar consonants; the claim is based on the imprecise usage of "palatal" to mean "palato-alveolar".

References

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2. Zsiga, Elizabeth C. (2013). *The sounds of language: an introduction to phonetics and phonology*. Chichester: Wiley-Blackwell. ISBN 978-1-4051-9103-6. OCLC 799024997 (https://search.worldcat.org/oclc/799024997).

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5. Titze, Ingo R. (January 2008). "The Human Instrument" (https://www.scientificamerican.com/article/the-human-instrument). *Scientific American*. **298** (1): 94–101. Bibcode:2008SciAm.298a..94T (https://ui.adsabs.harvard.edu/abs/2008SciAm.298a..94T). doi:10.1038/scientificamerican0108-94 (https://doi.org/10.1038%2Fscientificamerican0108-94). ISSN 0036-8733 (https://search.worldcat.org/issn/0036-8733). PMID 18225701 (https://pubmed.ncbi.nlm.nih.gov/18225701).

External links

- **Interactive places and manners of articulation** (http://www.chass.utoronto.ca/~danhall/phonetics/sammy.html) Archived (https://web.archive.org/web/20071220011219/http://www.chass.utoronto.ca/~danhall/phonetics/sammy.html) 2007-12-20 at the **Wayback Machine**

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